

2024 July 19

Streamlining Programmatic Access to Structural Biology Data with Python

RCSB PDB
info@rcsb.org

Habiba Morsy

Ivana Truong, Dennis Piehl, Brinda Vallat,
Christine Zardecki, Stephen K. Burley

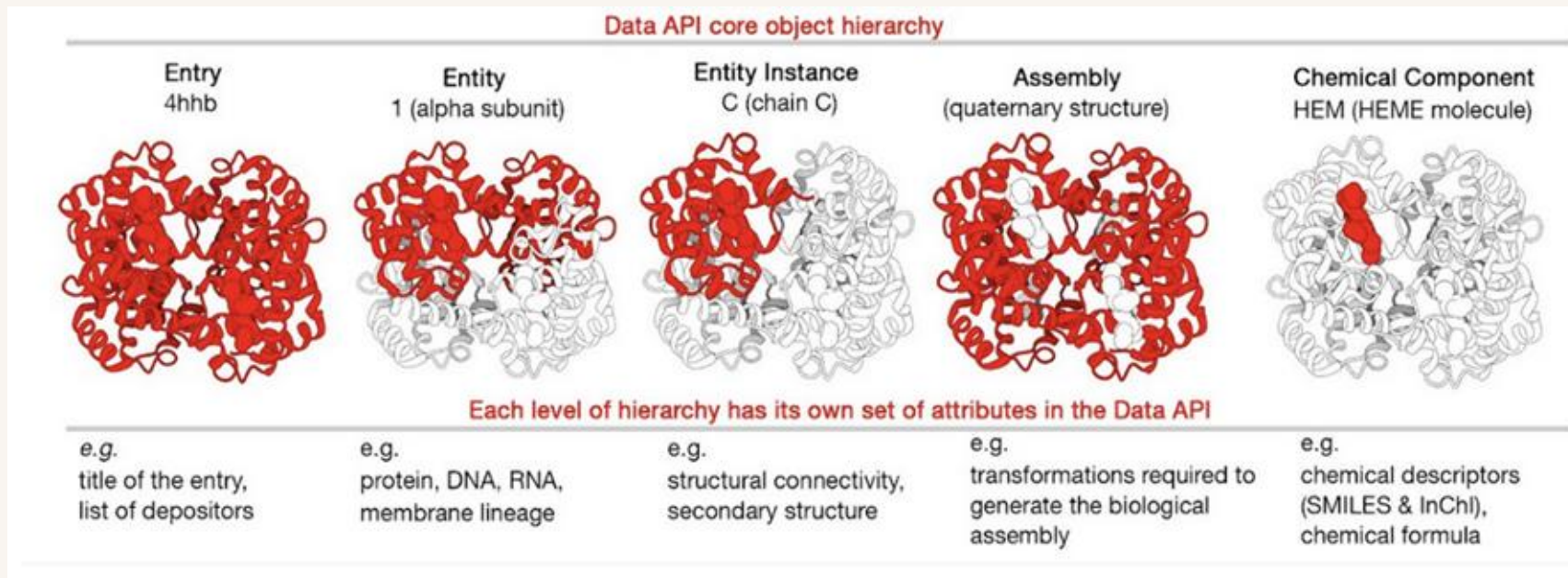
RCSB Protein Data Bank

- **PDB**: 1st open access digital data resource in biology
- Single global **archive** for 3D macromolecular structure data
- Over **222,000** structures of proteins, DNAs, and RNAs
- Serves many **millions** of users worldwide

rcsb.org is powered by publicly available APIs

PDB Data Organization

Hemoglobin: a protein in red blood cells that carries oxygen from the lungs to the rest of the body



Identifiers
“*rcsb_id*”

4HHB

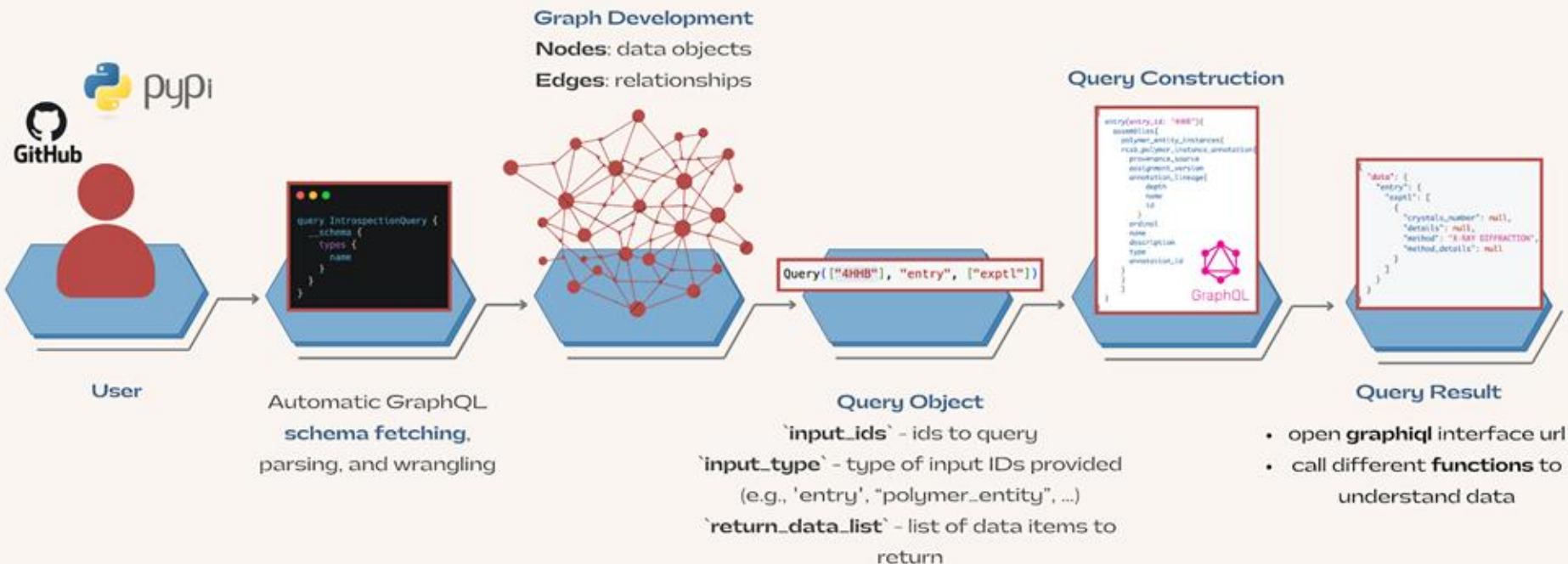
4HHB_1

4HHB.C

4HHB-1

HEM

Python Package for Data API



PDB Data Application Programming Interface (API)



GraphQL

```
{
  entry(entry_id: "4HHB"){
    exptl{
      method
      method_details
      crystals_number
      details
    }
  }
}
```

old input

json response

```
{
  "data": {
    "entry": {
      "exptl": [
        {
          "method": "X-RAY DIFFRACTION",
          "method_details": null,
          "crystals_number": null,
          "details": null
        }
      ]
    }
  }
}
```

rcsb.org UI result

Response

Request



Python Package



Database



New Input

```
Query(["4HHB"], "entry", ["exptl"])
```